

Predictive Analysis for Suicide Ideation Detection by Using Natural Language Processing and Machine Learning algorithms

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Abstract—Suicide represents a serious, yet extremely inescapable issue of public health concern [2]. Conventional approaches to suicidal ideation are extremely reliant on clinical screening, which is often limited by real-world constraints on scalability [5]. The growing usage of social media means that one often has suicidal ideation expressed using unstructured text on those platforms [1]. In this context, there is a proposition for using combined Natural Language Processing & Machine Learning for suicidal ideation detection, earlier than its onset. This model uses a balanced dataset comprising 2,32,074 posts on Reddit platform to model classical classifiers such as, logistic regression, K-Nearest Neighbours, Random Forest, using GloVe embeddings - and those using a transformer model known by BERT. Classical models usually support a maximum of 87.12% accuracy, while BERT performs better with an accuracy rating of 98.49% and an AUC score of 0.9988. These clearly establish that transformer-based models efficiently perform better on contextual embeddings regarding emotions, meaning, bringing into focus their compatibility and utility potentials for mental screening practices. Differences can still be resolved by enhancing sampling, observations, or strategies meant for imbalance data handling [5], [11], [12].

Index Terms—Suicide Ideation Detection, Natural Language Processing, BERT, Transformer Models, Machine Learning, GloVe embedding, Sentiment Analysis, Logistic Regression, Random Forest, K-Nearest Neighbor.

I. INTRODUCTION

Despite global healthcare systems significant advancement, suicide still remains a major cause of preventable mortality around the world with over 47,000 annual deaths in the US alone alongside high rates of anxiety (28.2%) and depression (24.4%) symptoms [11]. Early detection of suicidal ideation is critical for successful intervention; however, traditional clinical evaluation methods are mostly reactive and heavily reliant on self-disclosure, which restricts their ability to effectively predict [2], [5], [12]. Numerous studies shows that one of the primary challenges in suicide prevention and detection is the delay between ideation onset and clinical intervention. According to the comparative studies available, these types of social media-based detection systems can help alleviate this waiting period to a great extent by monitoring different linguistic patterns like the usage of "I" to refer to oneself, negative words, and feelings of hopelessness and despair [2]. Also,

with the widespread usage of different social media platforms available today, there is an ample amount of unorganized text data available that has been posted by users on these platforms to express their emotional pain and despair of self-harming thoughts. Specifically, the Reddit platform is actually an ideal setting in which such a suicidal tendency can be investigated given that it has an anonymous platform which minimizes stigma and enables users in giving adequate descriptions of their thoughts [12]. Such unorganized data sources give ample chances of being analyzed by NLP and Machine Learning models and methodologies to make predictions accordingly [1], [11].

II. PROBLEM STATEMENT

The paper focuses on exploring the use of Unstructured Text Analysis on social media platforms and how machine learning and NLP technologies can ensure authentic early prediction of suicide intentions. We highlight some of the shortfalls of traditional methods involving word vector representations in attaining implicit emotional expressions and indications not explicitly stated in words. To meet this objective, we propose this research to identify signs of suicidal tendency and indication through linguistic patterns and deep-embedded meanings in social platforms. Some research compare machine learning models with deep transformer models to determine their effectiveness on GloVe and BERT models to identify possible risks of suicide to ensure early intervention. Furthermore, based on a comparative evaluation performed in past research efforts, this study presents a mix of models to enhance accuracy and effectiveness within traditional methods to eliminate noisy patterns and models [2]. The entire procedural flow including data intake and processing, exploratory data analysis, and development of GloVe and BERT models has been implemented.

III. LITERATURE REVIEW

Currently emerging trends for suicide detection research reveal a shift towards rule-based systems and data-driven NLP techniques [12]. Traditionally, the techniques used a lot of keyword identification and sentiment analysis using

the lexicon approach, which lacked contextual understanding and were not robust enough to interpret the meaning. Deep learning techniques are more accurate than the traditional classifier because they are capable of learning the underlying linguistic form and implications of suicidal thoughts that fall into modifiable and nonmodifiable categories [2], [9]. Recent research has documented the efficacy of using machine learning classification methods in identifying mental health problems, using accuracy ranges from 78% to 84% [3], [6], [7].

BERT and its variants have proved their performance capability for modeling long-range dependency and context of user emotion in their stories, and they have been successfully integrated with techniques for suicide risk prediction [7]. According to a study by Li et al., the performance of transformer-based models like BERT surpasses that of traditional machine learning algorithms because of its capability to leverage the relationships between contexts and the dependency among texts [5]. It has been proved by a study conducted by Haque et al. that the usage of word embeddings increases the effectiveness of models and results in improved accuracy [2]. Ehtemam et al. proved that by the time accuracy of neural networks reached 0.70, the accuracy of Random Forest and XGBoost classifiers reached as high as 0.94 for the prediction of suicide risk assessment, while the maximum AUC for XGBoost classifiers reached 0.97 for the prediction of suicide risk assessment [1]. Again, for depression assessment on clinical interviews, the accuracy of up to 84% for Random Forest and 82% for XGBoost has been achieved by these models [6]. Deep learning models like BiLSTM models prove to be quite effective because of their ability to track word flow in posts with higher verbosity. Hybrid approaches, which mix deep learning with machine learning classifiers, also demonstrate effectiveness in suicide ideation detection [7]. Although these models can provide greater accuracy to a model, there can be issues with them like imbalanced datasets, a shortage of accurate labels, or cross-platform effectiveness [9].

Other studies showed that NLP can be combined with Electronic Health Records (EHRs) and that unstructured data in clinical notes are an important source of antecedent indicators of suicidality. Vance et al. showed that models using NLP can successfully identify anhedonia and suicidal ideation in the clinical records of patients with Major Depressive Disorder and that the positive predictive value of the model for anhedonia and suicidal ideation with intent or plan was 0.98 and 0.93 respectively [10]. More contemporary literature reviews of Large Language Models also underscore their strength in inferring intricate states of emotions and psychological conditions, such as anhedonia, hopelessness, and emotional detachment, which serve as vital elements of suicidal intention. Analysis on high-risk posts showed certain linguistic features, such as expressions of hopelessness "never", "impossible", "I'm tired", guilt "sorry", fear "scream", "feel lonely", and extensive use of question marks to denote hesitation and anguish [9]. This becomes possible through the implementation of semantic-based models instead of merely polarity-based models [11].

Further discussion also asserts the significance of social media with respect to mental health monitoring [4]. The language-based signs in the Twitter world and found language shift to serve as an anticipatory sign of suicidal behavior. In a related insight, Wright-Berryman et al. validated machine learning models for identifying depression, anxiety, and suicide risk from open-ended clinical interviews, achieving AUCs of 0.77, 0.74, and 0.70, respectively, showing the feasibility of using virtual platforms to simultaneously screen for mental health [11].

IV. METHODOLOGY

It includes all the functional modules developed with respect to the loading, preprocessing of the data obtained from a dataset collected from Kaggle [4] using SuicideWatch and Reddit [8], tasks associated to eliminate noise to better allow the data to be processed for both classical and transformer models. Exploratory data analysis revealed some disparity in the number of post Characters, usage of words, and the sentiment of the two categories of data. A sentiment score was computed to aid in the classification of the two categories. The dataset was then split for testing and training the models. Some efficient classical machine learning models were trained with GloVe embeddings, while the fine-tuned BERT model was trained to detect greater underlying data in the posts. Initial results indicate the feasibility of both classical and BERT models to aptly identify suicidal posts to detect suicide ideations early.

A. Dataset Description

This study is based on data taken from the openly available Kaggle repository [4]. It consists of 2,32,074 anonymized textual posts gathered from the Reddit social network, divided into two classes: suicidal and non-suicidal posts. The posts were taken from discussions related to mental health, where users express their thoughts in a narrative way; besides, it was guaranteed that persons posting were anonymous, which grants the material to be appropriate for conducting research on the detection of suicide ideation by using machine learning methods.

This dataset was originally gathered from the subreddits SuicideWatch and teenagers of Reddit through the Pushshift API and was introduced for the first time in a previous work related to the detection of suicide ideation in social media forums [8]. The dataset is perfectly balanced, as there are 1,16,037 samples in each class, which avoids class imbalance. The correct class balance was assured, maintaining the integrity of the classes while verifying the distribution during the data-loading phase.

B. Text processing and cleaning

Raw reddit posts contain substantial noise, such as URLs, emojis, punctuation, and inconsistent formatting. To handle this a preprocessing pipeline was implemented to normalize the textual data. The steps include : Conversion to lowercase,

Removal of URLs, HTML tags, punctuation, digits, and special characters using regular expressions, Removal of excessive whitespace, Token-level normalization for BERT using pre-trained tokenizer, Label Encoding such as, binary: suicide or non-suicide, and padding and truncation for transformer input overall, reduced impact of commonly used words and assigning higher weights to important words. These preprocessing steps ensure uniformity across all models and improve the quality of the features [5].

C. Text Representation and Tokenization

GloVe Embedding Construction: For traditional machine learning models, the posts were transformed into numerical vectors using 100-dimensional GloVe embeddings. Each of the word tokens used, was linked to the corresponding pre-trained embedding. Moreover, the document embeddings were calculated using the average of the word embeddings in a post. This was done to allow the semantic connection but with keeping computational efficiency due to use of smaller resources. may be replaced by ignoring his presence without affecting the meaning of the speech.

BERT Tokenization (Transformer Model): The texts are preprocessed for the deep learning approach by the Word-Piece tokenizer algorithm in the BERT model. It includes the following: Splitting rare and less frequent words into smaller components, Adding [CLS] and [SEP] special tokens, Attention masks creation, Trimming or padding in the text features to a certain maximum length. These above steps, known as tokenization, allow the text to be properly processed through a self-attention mechanism in the BERT model.

D. Exploratory Data Analysis (EDA)

Word Distribution in the Posts: The suicidal posts were indeed long, consisting of an average of 202.34 words, while the rest or the non-suicidal posts had only an average of 60.67 words.

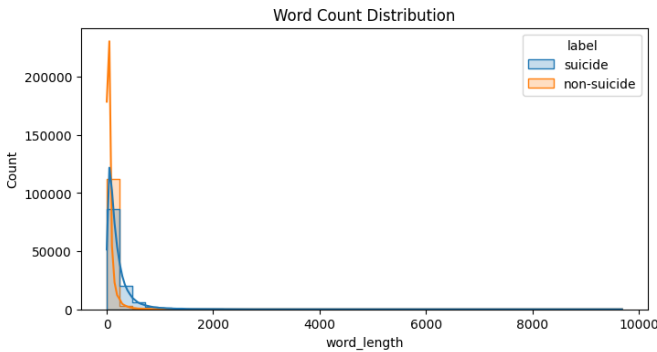


Fig. 1: Word count distribution

The histogram in Fig. 1 shows the skewed distribution of suicides, where the majority of the people in such drastic circumstances resort to the habit of narrating their viewpoints in lengthy ways. This corroborates the results of the psychological study, where the distressed individuals are more

inclined to feel, think, and express their ideas about the circumstances in depth, making the length of the post one of the major points of identification [1].

Token Frequency Analysis:

Haque et al. examined unigram frequencies in their Twitter data and discovered that suicidal tweets appeared with explicit negative emotion words such as "hate", "pain", indications of exhaustion "I'm tired", perceived worthlessness, and negation terms "don't want", "never", "nothing". Death-related terms also appeared commonly, such as "death", "want die", and "kill" whereas, non-suicidal tweets included words that showed happiness, positive attitudes "I'm happy", "want fun", "laugh loud", and social engagement "job", "work", "get better" [2]. From Reddit data, not only was the disturbance in their mind illustrated through the use of the most frequently used words in the posts on suicides, but the emotional suffering they felt was also evident through the use of words like "im", "ive", "don't", "feel", "life", "think", and "want". These lexical features provide viable indicators that are present consistently in both the Twitter tweets and the Reddit comments, thereby showing that word choice is an effective discriminative cue when it comes to suicidal ideation in text. Thus, our approach for feature engineering gives prime importance to these lexical and sentiment features for their impact on model performance.

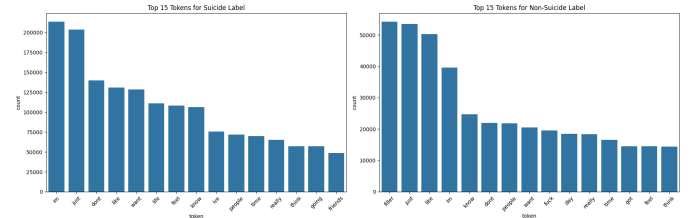


Fig. 2: Token frequency analysis

The token frequency bar chart in Fig. 2 not only effectively displays the use of the first-person pronouns and emotional verbs in the posts relating to suicide, but also the linguistic phenomenon of the use of first-person pronouns and emotional verbs also symbolizes the tendency towards increased self-focus and emotionality, which are already taken as strong predictors of suicide in psychological and computational analysis [11].

E. Feature Engineering of Sentiment and Lexical

A Sentiment Balance Score was designed based on predefined positive and negative lexical cues. A Sentiment Balance Score indicates that emotional polarity always overrides even the indication of having a positive or negative sentiment label.

Mean score (suicidal ideation): -0.3196

Mean score (Non-Suicidal): +0.2077

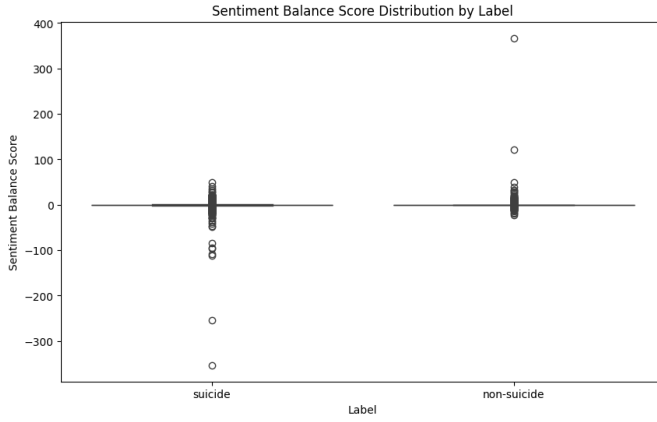


Fig. 3: Sentiment balance score distribution

The plotting in the Fig. 3 figure shows clear bifurcation between the two classes, which hints that posts indicating suicide tendency lean toward negative sentiment. There is a slight overlap between the two distributions, affirming that the polarity of sentiment is perhaps the most distinguishing feature and hence its inclusion in classical machine learning models is fully justified.

F. Train-Test Split

The split of the dataset into the training and testing sets was performed in an 80:20 ratio, with the class balance being maintained for the split. This was accomplished using randomly that was set in such a way that the results are reproducible.

G. Classical Machine Learning Models

GloVe embeddings were used to train three classic classifiers: Logistic Regression, Random Forest, K-Nearest Neighbors

1) Random Forest (RF):

Random Forest is an ensemble learning method that constructs multiple decision trees at training and then aggregates their predictions through voting in the case of classification or averaging in the case of regression to come up with a final prediction. It uses bootstrap aggregating, where each tree is trained on a random subset of the training data sampled with replacement. The average predictions over a large number of trees reduce overfitting and variance significantly compared to a single decision tree, with low bias. It works well with high-dimensional data, handles non-linear relationships naturally, is very robust against noisy data and outliers, and provides by default the importance of features.

2) Logistic Regression (LR):

Logistic Regression is a classification algorithm that predicts the probability of an input belonging to a specific class by applying a sigmoid function to a linear combination of features, producing outputs between 0 and 1. Key advantages include avoiding overfitting, computational efficiency, interpretability, excellent scalability to large datasets, and natural probabilistic outputs that provide accurate estimates. Limitations largely revolve around its assumption of linear

relationships between features and log-odds of outcomes; it may not capture complex non-linear patterns without feature engineering. Extensions to include multinomial logistic regression allow for multi-class problems, making it a versatile and widely-used baseline model across many different classification tasks.

3) K-Nearest Neighbors (KNN):

The K-Nearest Neighbors is an instance-based Machine Learning algorithm used for creating predictions by finding the K-most similar training instances for the data point and voting based on their training labels, either by majority votes for classification tasks or averaging for regression tasks. The main problem associated with K-Nearest Neighbors is it has high computation costs during prediction for large-sized datasets, and limited dimensionality, dependency on the scaling of the features as it considers large-valued features for computation of the nearest neighbors for the dataset.

H. Transformer-Based Model: Fine-Tuned BERT

The GloVe embeddings were used to train three classic classifiers: Logistic Regression, Random Forest, K-Nearest Neighbors. BERT-base was finetuned for binary classification, adding only a dense classification head on top of the output of the CLS token.

Technical details synchronized with the implementation:

Pre-trained transformer encoder, Fine-tuning end-to-end, Loss function of cross-entropy, Optimization of AdamW, Learning rate scheduling, Mini-batch training.

I. Model Selection Rationale and Training Procedure

Models selected from this scenario were driven by two major reasons: (i) to create effective and significant baseline models by applying classical machine learning approaches, and (ii) to evaluate how efficiently the state-of-the-art transformer model is able to learn complex contexts and emotional aspects linked to suicidal thoughts. To implement classical models, grid search strategies were adopted to define the values of the hyperparameters while training the models. On the other hand, early stopping was adopted to avoid overfitting while training the BERT model. Validation of the model necessarily has to be applied on the test data.

V. RESULTS AND EVALUATION

A. Quantitative Results

TABLE I: Performance Comparison of Models

No.	Model	Accuracy	Precision	Recall	F1-score	ROC AUC
1	Logistic Regression	0.860196	0.834654	0.898350	0.865332	0.920218
2	Random Forest	0.871270	0.897784	0.837937	0.866829	0.943773
3	K-Nearest Neighbors	0.790908	0.715880	0.964666	0.821858	0.890185
4	BERT	0.984940	0.987482	0.982333	0.984901	0.998727

The accuracy, precision, recall, F1-score, and AUC-ROC values were employed as criteria for the classification of the models. The conventional models based on machine learning, which employed Glove embeddings, performed well but had

difficulty interpreting the complex context. Results found from Classical ML models:

- The highest accuracy of 87.12% using the traditional models for the Random Forest algorithm was obtained, with precision of 89.77%, recall of 83.79%, F1-score of 86.68%, and AUC of 0.943.
- With an accuracy of 86.01%, precision of 83.46%, recall of 89.83%, F1-score of 86.53%, and AUC of 0.920, Logistic Regression came next.
- The results of K-Nearest Neighbours were 79.09% accuracy, 71.58% precision, 96.46% recall, 82.18% F1-score, and 0.890 AUC.

The improved BERT model, on the other hand, performed better on every metric from ML models: accuracy 98.49%, precision 98.74%, recall 98.23%, F1-score 98.49%, and AUC 0.9987.

With only 1.51% misclassifications as compared to 12.88% for Random Forest, confusion matrices showed that BERT had far fewer false negatives, which are crucial for suicide detection. These findings highlight BERT's near-perfect AUC, which demonstrates its capacity to identify insignificant linguistic indicators. Therefore, The study reports that machine learning models were able to successfully and efficiently identify suicide risk from open-ended language while performance exceeded chance-level prediction.

B. Performance Metrics

Confusion Matrix Based Evaluation:

A comparison of the predicted labels and actual labels of the test data led to the adoption of a confusion matrix in Fig. 4 to assess the performance of both the optimized BERT model and traditional machine learning algorithms, allowing for a thorough examination of correct and erroneous classifications of posts as suicidal and non-suicidal. The confusion matrix was used to measure true positives (TP), which denote correct suicidal posts; true negatives (TN), which denote correct identifications of non-suicidal posts; false positives (FP), which denote erroneous identifications of non-suicidal posts as suicidal; and false negatives (FN), which denote erroneous identifications of suicidal posts as non-suicidal.

An assessment of the outcomes indicated that although traditional machine learning algorithms systematically predicted suicidal content reasonably (TN was high), they tended to have higher false negatives, indicating that suicidal posts were overlooked. This is more problematic because, among other things, high-risk suicidal posts went undetected, posing safety concerns. Contrarily, the BERT model showed a good balance between TP and TN with relatively fewer FPs and FNs, indicating more reliable detection of suicidal content with a minimum number of unnecessary false alarms. Further illustration was done through confusion matrix visualization for performance characteristics, class-wise behavior, and prediction errors.

Accuracy, precision, recall, and F1-score are standard performance metrics computed using values from the confusion matrix to enable a fair and consistent comparison across

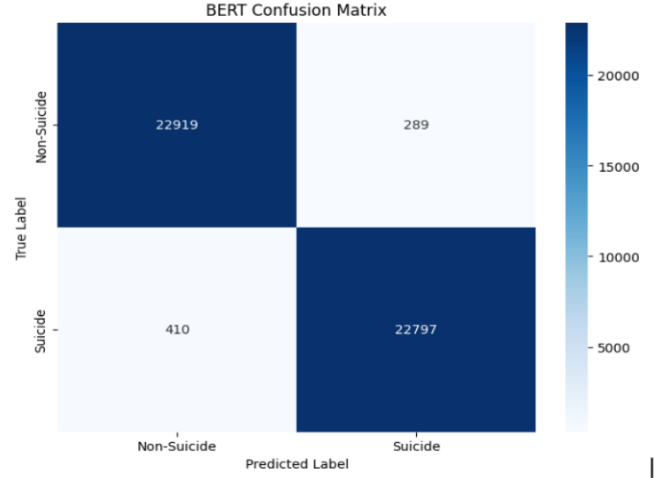


Fig. 4: BERT confusion matrix

all models. In this light, this study's confirmation of higher performance by the transformer-based approach than by the classical methods, especially in detecting suicidal posts while maintaining high accuracy for non-suicidal content, was based on the confusion matrix-based evaluation.

VI. DISCUSSION

This research shows the application Natural Language Processing to process the texts, employed machine learning classifiers for suicide ideation detection also analyzed linguistic and semantic features from spoken responses. Recent studies have proved the success of deep learning ensemble methods over standalone models in identifying suicidal ideations on social media platforms [9]. Some encouraging approaches have also been observed by combining transformers with traditional classifiers [7]. The use of semantic embeddings and sentiment features enhances the performance of classical machine learning models, but these models still struggle to interpret the meaning of the context. BERT, though, pulls far ahead of the models for using technology of contextualized embeddings and the application of a bidirectional attention mechanism. The enhanced performance of BERT compared with traditional approaches has been compared in a study performed by Li et al., indicating that ML approaches applied on transcripts of clinical interviews could predict the risk of suicide with AUC values from 0.76 to 0.92 without direct disclosure of suicide ideation [5]. However, because they are easier to understand and need less processing power, classical models are still useful. Models were able to efficiently detect suicidal ideation from textual response. NLP models successfully detected suicidal ideation and anhedonia from text.

VII. LIMITATIONS AND ETHICAL CONSIDERATIONS

Though our research efficiently helps in detecting early risk and predictions, still there are some matters that need to be

noted and looked into. Social media data does not equate to clinical diagnosis language and cultural bias in Reddit-based datasets. The possibility of false positives occurring through automated systems. The need for human-in-the-loop deployment. Although our research relied on the Social platform data, Wright-Berryman et al., Vance et al., in fact, proved the efficiency of such NLP techniques in clinical environments by means of structured interviews as well as EHRs [10], [11]. Furthermore, the dataset's limited generalizability to non-English environments emerges from its emphasis on English-language posts. Ethical considerations regarding automated risk flagging needs to be noticed. Thorough ethical reviews are necessary to address any biases in annotation and the possibility that model implementation might raise misunderstanding.

VIII. CONCLUSION AND FUTURE WORK

This study evidences that transformer-based models, specifically BERT, which captures contextual, emotional, and semantic patterns, are able to outperform traditional machine learning classifiers like Logistic Regression, Random Forest, and KNN in detecting suicidal ideation from social media texts. Among them, the highest accuracy reached was 87.12%, while the best fine-tuned BERT model achieved 98.49% on the dataset of 232,074 posts on Reddit, with an AUC of 0.9988, demonstrating strong potential for practical early detection systems, transformer-based contextual embeddings can pick up subtle emotional and semantic cues in text, which classical models lack, hence being suitable for real-time monitoring of online platforms. Future work can extend this by developing the approach further onto multilingual datasets, larger and more diversified populations, explainable AI methods, advanced NLP techniques, and prospective clinical validation with a view to ensuring ethical, privacy-preserving, and real-world applicability in mental health monitoring. Further refinement of such systems may position them as support tools for clinicians that enable timely intervention and improve suicide prevention at scale.

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